

Adjoint-State Gradients on the Stratified Voxel Grid

Waveform-inversion gradients through the Foldy–Lax solve,
and their Dietrich–Kormendi limit

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Abstract

The two-potential formulation of this programme — a depth-stratified reference solved exactly, plus a lateral contrast of cubic voxels summed by a Foldy–Lax solve carried by the reference Green’s tensor — is the natural forward model for waveform inversion. Its first-order term *is* the Dietrich–Kormendi (1990) Born perturbation of the plane-wave reflectivity; the full resolvent is its multiple-scattering generalisation. This note derives the adjoint-state gradient of a waveform misfit with respect to every voxel’s contrast. The gradient costs one extra Krylov solve per source, frequency and transverse wavenumber, independent of the number of voxels, and the adjoint system is the *forward* system at $-k_y$, conjugated by $\mathbf{W}_\pm = \mathbf{I}_\pm \mathbf{W}$, the Voigt weight times a displacement–strain parity — the nine-component form of the thesis identity $\mathcal{H}^\dagger(k_y) = \mathcal{H}(-k_y)$. That identity requires matching reciprocity laws of both G_0 and T_0 , and both are measured here. The G_0 law holds on the assembled sweep operator to 10^{-13} or better, in a whole space and in a stratified background, and the Foldy–Lax identity to 10^{-16} (gate A1). The T_0 law is exact for the Rayleigh cube carried by the sweep solver, and *failed* by the resonance composite, which is incidence-dependent and must not be used as it stands. A1 also corrected this note’s first draft, which stated the law with $\mathbf{W}$ alone and the Foldy–Lax operator in the wrong order. The gradient is implemented (`sweep_gradient.py`) and passes a six-rung validation ladder (§9). That covers the receiver transpose and the T_0 derivative (complex step is invalid, since T_0 is not holomorphic in its contrast), an absolute check of the Born limit against the exact thin-layer reflection, a dot-product test, and a Taylor test against the nonlinear misfit. The last two reach spectral radius 0.74–0.80.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T -matrix programme $T = T_0(I - G_0 T_0)^{-1}$: (I) the stratified inter-site propagator (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the seismic grid T -matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); (IV) adjoint-state gradients through that solve, for waveform inversion (`AdjointStateGradients`). This is note (IV).*

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1 What already exists

Nothing below is new machinery until §3. The inversion problem sits on four things this programme and its sibling repository have already built, and one thing the thesis already proved.

Object	Where	Validated to
Forward operator ($\mathbf{I} - G_0 T_0$), 2½-D, stratified G_0	note (III); <code>sweep_solver</code> , <code>directional_sweeps</code>	GMRES = LU = Neumann 1.2×10^{-12} ; ΔG_0 vs Kennett 2×10^{-10}
Forward operator, 3-D	note (III) §“three-dimensional case”: real-space tables, reverberation spectral	designed and costed (2.1 h build at 16^3); not gated end to end
Composed $P^z + P^x$ on the slab kernel	<code>scripts/composed_matvec.py</code>	<i>blocked</i> : GATE 1 fails, lattice aliasing. Not needed here
Adjoint of the Foldy–Lax operator	thesis <code>VariationalSum.tex</code> , Eqs. <code>sym</code> , <code>HsymDef</code> , <code>Htdef</code>	proved, in the thesis’s own representation
1-D Fréchet derivatives, = Dietrich–Kormendi	<code>SeismicInversion/latex/ frechet_derivatives.tex</code> §“Equivalence to D&K”	AD vs tangent-linear $< 10^{-10}$; vs finite differences $< 10^{-5}$
1-D Levenberg–Marquardt, Hessian hierarchy	same, §§“Hessian”, “Global Matrix Method”, “LM inversion”	gated in <code>Kennett_Reflectivity</code> , <code>GlobalMatrix</code>

Two consequences shape the rest of this note. First, the adjoint is built on the *sweep* operator of note (III), not on the composed slab matvec, whose blocker is a representation problem the sweeps do not have. Second, the reference-layer derivatives are a solved problem in one dimension; what is missing is the gradient with respect to the *voxel* contrasts, through the multiple-scattering solve. That is the subject here.

2 Forward model and misfit

Fix a source s , frequency ω and, in 2½-D, a transverse wavenumber node k_y . The nine-component exciting field ψ on the voxel grid solves

$$(\mathbf{I} - G_0 T_0) \psi = \psi^{\text{inc}}, \quad (1)$$

with $T_0 = \text{blockdiag}_i T_0^{(i)}(m_i)$ the per-voxel cube T-matrices and $m_i = (\Delta\lambda_i, \Delta\mu_i, \Delta\rho_i)$ the contrast of voxel i from the local reference. The data are the reference response plus the field

radiated by the contrast sources $\tau = T_0\psi$,

$$d = d^{\text{ref}} + \mathbf{R} T_0 \psi, \quad (2)$$

where $\mathbf{R}$ carries the stratified Green's tensor from each voxel to the receivers (the $\mathcal{P}_s^{z,R}$ of the thesis `LSstrat`). With residual $r = d - d^{\text{obs}}$,

$$\mathbf{M} = \frac{1}{2} \sum_{s,\omega,k_y} w_{k_y} r^{\text{H}} r. \quad (3)$$

Moduli are complex (constant- Q), so every adjoint below is a *transpose*, never a Hermitian conjugate: the operators are complex-symmetric in a weighted sense, not Hermitian. The only conjugate in the problem is the one on the residual.

3 The voxel gradient

Perturb $T_0 \rightarrow T_0 + \delta T_0$. From (1), $\delta\psi = (\mathbf{I} - G_0 T_0)^{-1} G_0 \delta T_0 \psi$, and so

$$\delta d = \mathbf{R} [\mathbf{I} + T_0 (\mathbf{I} - G_0 T_0)^{-1} G_0] \delta T_0 \psi = \mathbf{R} (\mathbf{I} - T_0 G_0)^{-1} \delta T_0 \psi, \quad (4)$$

by the push-through identity $\mathbf{I} + A(\mathbf{I} - BA)^{-1}B = (\mathbf{I} - AB)^{-1}$. Define the adjoint field $\tilde{\psi}$ by

$$\boxed{(\mathbf{I} - G_0^{\text{T}} T_0^{\text{T}}) \tilde{\psi} = \tilde{\mathbf{b}}, \quad \tilde{\mathbf{b}} = \mathbf{R}^{\text{T}} \bar{r}} \quad (5)$$

so that $\bar{r}^{\text{T}} \mathbf{R} (\mathbf{I} - T_0 G_0)^{-1} = \tilde{\psi}^{\text{T}}$. The tilde follows the thesis (`VariationalSum.tex`, Eqs. `bvardef` and `btdef`). There the incident field $\mathbf{b}$ is paired with $\tilde{\mathbf{b}} = [\mathcal{P}_s^{z,R}]^{\text{T}} \tilde{\mathbf{f}}$, radiated from the receiver with directivity $\tilde{\mathbf{f}}$. The right-hand side of (5) is that field with $\tilde{\mathbf{f}} = \bar{r}$, and $\tilde{\psi}$ is its exciting field. Then $\delta\mathbf{M} = \text{Re} \sum \tilde{\psi}^{\text{T}} \delta T_0 \psi$, and because T_0 is block-diagonal the gradient is a local 9×9 contraction at each voxel:

$$\boxed{\frac{\partial \mathbf{M}}{\partial m_{i,\ell}} = \text{Re} \sum_{s,\omega,k_y} w_{k_y} \tilde{\psi}_i^{\text{T}} \frac{\partial T_0^{(i)}}{\partial m_{i,\ell}} \psi_i, \quad \ell \in \{\Delta\lambda, \Delta\mu, \Delta\rho\}. \quad (6)}$$

The right-hand side of (5) is the residual injected at the receivers and carried to the voxels by the transposed receiver operator. By reciprocity $\mathbf{R}^{\text{T}}$ is the reference field radiated *from* the receivers, with the same weight $\mathbf{W}_{\pm}$ as §4: $G(r \leftarrow v)^{\text{T}} = \mathbf{W}_{\pm} G(v \leftarrow r) \mathbf{W}_{\pm}^{-1}$. For a receiver recording displacement, $\mathbf{R} = \Pi_u G(r \leftarrow v)$ with Π_u selecting the three displacement rows, and $\mathbf{W}_{\pm}^{-1} \Pi_u^{\text{T}} = \Pi_u^{\text{T}}$, so

$$\mathbf{R}^{\text{T}} \bar{r} = \mathbf{W}_{\pm} \cdot [\text{field at the voxels from a point force } \bar{r} \text{ at the receiver}], \quad (7)$$

which the existing layered incident-field code already computes. Both statements are measured in §4.1. In 2½-D the observed data live in real y , so the residual enters node k_y with the phase and weight of the k_y quadrature, $w_{k_y} e^{ik_y y_r} \bar{r}(y_r)$.

Status. Equations (5)–(6) are an exact algebraic identity for any G_0 and T_0 ; nothing about the physics is assumed. Their implementation passes the dot-product and Taylor tests A4 and A5 (§8.1, §8.2).

The derivative of the cube T-matrix. $\partial T_0/\partial m$ is not the Born weight. The Rayleigh cube carries the *effective* contrasts $(\Delta\rho^*, \Delta\lambda^*, \Delta\mu_{\text{diag}}^*, \Delta\mu_{\text{off}}^*)$, which are nonlinear in $(\Delta\lambda, \Delta\mu, \Delta\rho)$ through the Eshelby amplification factors (`effective_contrasts._compute_amplification_factors`). That is the point: the gradient is exact for the resummed intra-voxel scattering, which a Born kernel is not.

The first draft of this note proposed a complex-step derivative. That is invalid here. Complex step, and its generalisation the Cauchy contour integral, require T_0 to be *holomorphic* in m , and it is not: `compute_cube_tmatrix` applies its real form factors to the *real parts* of the effective contrasts only ($\Delta\mu^* \mapsto \text{Re } \Delta\mu^* f_\mu + i \text{Im } \Delta\mu^*$, and likewise for κ and ρ). Every contrast-dependent ingredient is otherwise a smooth rational function, so T_0 is smooth as a map of *real* m , away from the resonance poles of the amplification factors. The derivative is therefore taken by Richardson-extrapolated central differences along the real axis. There are three parameters per voxel type, so it costs nine T_0 evaluations. At zero contrast the amplification factors are exactly 1, and the derivative reduces to the Born weights $\omega^2 V \mathbf{I}_3 \oplus 0$ and $0 \oplus V \partial c_{\text{Voigt}}$ only as $ka \rightarrow 0$. At finite ka it carries the form factor, which the Born weight does not. Gate A2 measures all of this (§3.1).

3.1 The derivative, measured (gate A2)

`gate_a2_t0_derivative` uses the project’s validated test parameters ($\alpha = 5$, $\beta = 3$, $\rho = 2.5$; contrast (2 GPa, 1 GPa, 0.1 g/cm³); $a = 0.5$ km) at $ka_S = 0.05$ and 0.3, and at a complex frequency ($\omega(1 + 0.03i)$) as in the sweep gates.

Check	$ka_S = 0.05$	$ka_S = 0.3$	$ka_S = 0.3$, complex ω
Richardson error estimate, worst parameter	2.5×10^{-11}	3.4×10^{-13}	2.9×10^{-13}
observed order of raw differences	2.00 (see text)	2.00	2.00
random direction vs $\sum_\ell u_\ell \partial_\ell T_0$	1.0×10^{-13}	1.0×10^{-13}	1.6×10^{-13}
<i>must fail</i> : imaginary-step derivative	$4\text{--}7 \times 10^{-4}$	$1.5\text{--}2.4 \times 10^{-2}$	$1.5\text{--}2.4 \times 10^{-2}$

The imaginary-step row is what complex step would have delivered. It is wrong by an amount that grows like $(ka)^2$, which is at least 10^7 times the finite-difference error. The density channel at $ka_S = 0.05$ is linear in $\Delta\rho$ to rounding, so its raw differences agree to rounding and have no truncation order to measure. The gate reports it as linear rather than asserting an order. (The first run asserted a fixed 10^{-3} for the imaginary-step disagreement and an order for every channel; both failed for those reasons, not for any defect in T_0 .)

Born limit. At zero contrast, the relative difference from the Born weights is:

ka_S	0.40	0.20	0.10	0.05
$\partial_{\Delta\lambda}, \partial_{\Delta\mu}, \partial_{\Delta\rho}$	1.90×10^{-2}	4.79×10^{-3}	1.20×10^{-3}	3.00×10^{-4}

The fitted slope is 2.00, 1.99, 1.99, which is the $(ka)^2$ convergence the form factor predicts. The residual is identical in all three channels at every ka , with a coefficient of about 0.119 in $(k_S a)^2$. That is consistent with a single contrast-independent geometric form factor at zero contrast. This is an observation; the code path that makes the three equal has not been traced.

4 The adjoint is the forward solve at $-k_y$

Equation (5) needs a solver for the transposed operator. It does not need a new one. Both factors obey one diagonal similarity law (measured below),

$$G_0(k_y)^\top = \mathbf{W}_\pm G_0(-k_y) \mathbf{W}_\pm^{-1}, \quad T_0^\top = \mathbf{W}_\pm T_0 \mathbf{W}_\pm^{-1}, \quad \mathbf{W}_\pm = \mathbf{I}_\pm \mathbf{W}, \quad (8)$$

$$\mathbf{W} = \text{diag}(1, 1, 1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}), \quad \mathbf{I}_{\pm} = \text{diag}(1, 1, 1, -1, -1, -1, -1, -1, -1).$$

$\mathbf{W}$ is the Voigt engineering factor. $\mathbf{I}_{\pm}$ is the parity between the displacement and strain halves of the state: the two off-diagonal blocks of the 9×9 carry one spatial derivative, taken at the receiver in one and at the source in the other, so exchanging source and receiver flips their sign. The adjoint system (5) is the transpose of $(\mathbf{I} - T_0 G_0)$, *not* of $(\mathbf{I} - G_0 T_0)$, and $G_0^T T_0^T = \mathbf{W}_{\pm} G_0(-k_y) T_0 \mathbf{W}_{\pm}^{-1}$, so

$$(\mathbf{I} - T_0 G_0(k_y))^T = \mathbf{W}_{\pm} (\mathbf{I} - G_0(-k_y) T_0) \mathbf{W}_{\pm}^{-1} \quad (9)$$

and (5) becomes a forward solve at $-k_y$ with the right-hand side $\mathbf{W}_{\pm}^{-1} \tilde{\mathbf{b}}$, solved for $\tilde{\psi}_{-}$, followed by $\tilde{\psi} = \mathbf{W}_{\pm} \tilde{\psi}_{-}$. This is the nine-component statement of the thesis result: with $\Delta \mathcal{C}_{\text{eff}}(k_y) = \Delta \mathcal{C}_{\text{eff}}^T(-k_y)$ and $\mathcal{P}_{r,s}(k_y) = \mathcal{P}_{r,s}^T(-k_y)$ (`VariationalSum.tex`, Eq. `sym`), the Lippmann–Schwinger operator $\mathcal{H}(k_y) = \mathbf{I} - \mathcal{P}_{r,s}(k_y) \Delta \mathcal{C}_{\text{eff}}(k_y)$ has adjoint $\mathcal{H}^{\dagger}(k_y) = \mathcal{H}(-k_y)$ (Eq. `Htdef`). The thesis used that identity to build Schwinger–Levine variational estimates; here it makes the gradient cost exactly one forward solve. Two practical consequences follow. The k_y grid of note (III) is symmetric about zero, so the $-k_y$ operator is already in the cache; and the forward and adjoint solves share an operator and can share a Krylov space. In 3-D, where k_y is not a parameter, the same law is a pair swap on the real-space tables, and the adjoint reuses the tables built for the forward solve.

The T_0 half, measured. `g:t0_reciprocity` checks $T_0 \mathbf{W}^{-1}$ for symmetry, i.e. $T_0^T = \mathbf{W} T_0 \mathbf{W}^{-1}$. A cube T_0 has no displacement–strain block, so $\mathbf{I}_{\pm}$ commutes with it, and this is the second law of (8):

T_0	ka_S	asymmetry of $T_0 \mathbf{W}^{-1}$	normal–shear block
Rayleigh cube	0.05, 0.1, 0.3	0	0
Resonance composite, $n_{\text{sub}} = 3$, $\hat{k} = \hat{z}$	0.5	1.6×10^{-2}	2×10^{-19}
Resonance composite, $n_{\text{sub}} = 3$, $\hat{k}$ oblique	1.2	1.7×10^{-1}	4×10^{-4}

The Rayleigh result is structural and was expected: the cube is block-diagonal, the density block is a multiple of $\mathbf{I}_3$, and a cubic Voigt stiffness has no normal–shear coupling, so $\mathbf{W}$ commutes with T_0 and the law reduces to plain symmetry. The gate’s value is as a regression guard, with the composite rows as its calibration leg: they show the check can fail.

The composite fails, and not by a rounding amount. It is $\sum_n T_{\text{loc}} \psi_n^{\text{exc}}$ for sub-cell exciting fields driven by one incident plane wave, so it carries the incident phase on its input side and none on its output side, and it changes with $\hat{k}$. It is a response to one illumination, not a field-independent operator, and *no* such object can be reciprocal. The adjoint field arrives from the receivers, from a different direction than the source field, so a gradient built on the composite would be wrong by the asymmetry in the table. Above the Rayleigh range the options are to keep the sub-cells as explicit unknowns in the global solve (so the local operator is T_{loc} , which is Rayleigh and reciprocal), or to rebuild the composite with a symmetric projection. The 27-component Galerkin T_0 has not been checked.

The G_0 half, measured on the assembled operator (gate A1). The pieces had been checked separately (the whole-space $\mathbf{W}$ -symmetry of `StratifiedComposition`; the J_6 law of the reverberation, note (III) [D3]), but not the assembled discrete sweep operator, which is what (9) applies. `gate_a1_sweep_reciprocity` materialises it densely at $\pm k_y$, split into the lateral sweep, the same-plane reverberation and the inter-plane coupling. Residuals are $\|A - B\|/\|A\|$ at the finer of two quadratures; every one is flat under refinement.

Claim	whole space	stratified
G_0 law (8), lateral sweep	2.8×10^{-14}	1.6×10^{-16}
same-plane reverberation	—	1.0×10^{-13}
inter-plane coupling	1.0×10^{-14}	5.1×10^{-16}
assembled G_0	2.1×10^{-14}	1.7×10^{-15}
identity (9), Rayleigh T_0	4.4×10^{-17}	3.0×10^{-16}
<i>must fail:</i> $\mathbf{W}$ in place of $\mathbf{W}_\pm$	0.74	0.29
$\mathbf{W}_\pm$ without the k_y flip	0.41	0.097
$(\mathbf{I} - G_0 T_0)^\top$ for $(\mathbf{I} - T_0 G_0)^\top$	2.4×10^{-3}	0.095

$\mathbf{W}_\pm$ was measured, not assumed. A free fit of $[G_0(k_y)^\top]_{ab}/[G_0(-k_y)]_{ab}$ is an exact outer product c_a/c_b , spread 1.4×10^{-13} (whole space) and 2.6×10^{-11} (stratified), with c equal in both to the diagonal of $\mathbf{W}_\pm$ given in (8). The whole space is at $\omega/2\pi = 1 + 0.03i$, $k_y = 0.6$, pitch 0.25; the stratified case is the fast-slab crust of the sweep-solver tests, planes 7 and 11, $\omega/2\pi = 6$, $k_y = 0.3$, $Q = 2$; both 2×4 voxels. $k_y \neq 0$ is essential, since at $k_y = 0$ the SH channel decouples and the flip is invisible.

Two errors this gate caught in the first draft of this note. The law was stated with $\mathbf{W}$ alone. That transcribed the whole-space $\mathbf{W}$ -symmetry, which compares entries at the same separation, as a transpose law, which reverses the separation; the reversal is where $\mathbf{I}_\pm$ comes from, and without it the law fails at 0.29–0.74. The identity (9) was also written for $(\mathbf{I} - G_0 T_0)^\top$; that operator does *not* map to the forward operator at $-k_y$ (third calibration row), and the adjoint system never needed it. Neither error would have been visible without the dense measurement: both drafts are algebraically self-consistent.

4.1 The receiver transpose, measured (gate A1b)

`gate_a1b_receiver_transpose` tests the law in 3-D, where there is no k_y to flip and it is a pure exchange of source and receiver. It checks the wavenumber-domain kernel `corrected_layered_9x9` first, then the real-space field of `layered_incident_field`. For the real-space check it builds the dense $\mathbf{R}$ one voxel at a time, with nine unit sources at each voxel read at the receiver, and compares it with the field radiated *from* the receiver. The model is the fast-slab crust of the sweep-solver tests, with voxel planes 7 and 11. One receiver is in the background above them (the same material as the voxels); the other is *inside* the slab (a different material), which is the case where taking the wrong interface’s material once left an 83% residual (`StratifiedComposition`).

Check	same material	cross material	$\mathbf{W}$ in place of $\mathbf{W}_\pm$
k -domain, $Q = 2, 1000$, surface off/on	$\leq 6.5 \times 10^{-16}$	$\leq 7.6 \times 10^{-16}$	0.30–0.40
real space, $k_r^{\max} = 10/\text{pitch}$ (documented rule)	1.2×10^{-16}	1.4×10^{-2}	0.30, 0.37
real space, $k_r^{\max} = 20/\text{pitch}$ (asserted)	9.6×10^{-17}	1.1×10^{-9}	0.30, 0.37
real space, $k_r^{\max} = 30/\text{pitch}$	9.6×10^{-17}	7.7×10^{-12}	0.30, 0.37
form (7), force at the receiver, 20/pitch	1.8×10^{-16}	2.6×10^{-10}	—

The node spacing is held at $\Delta k = 0.31$ while the cutoff grows. At $Q = 2$ the free surface changes nothing (it is damped out), so the free-surface leg is discriminating only at $Q = 1000$, which is why the k -domain check is run at both.

A defect in the transverse rule, outside the adjoint. The law holds; the documented rule does not resolve it. At $k_r^{\max} = 10/\text{pitch}$ the cross-material pair fails at 1.4×10^{-2} , flat in the node spacing (in a first run on a 3×2 lattice, 1.53×10^{-2} at 32 nodes and 1.52×10^{-2} at 64). Raising only the cutoff at fixed spacing brings it to 1.1×10^{-9} and then 7.7×10^{-12} .

So this is truncation, not a convention error, and it appears only when the two planes are in different materials. The construction subtracts the whole-space kernel *of the receiver plane's medium* spectrally and adds it back in closed form. When the media differ, what remains to be transformed is not reverberation alone but also the difference between the two direct paths, and `measure_dg0_transverse_rule` fixed the 10/pitch cutoff on a single pair of planes in the *same* medium (planes 51 and 50 of the `gate_sh_impedance` model). `layered_stack_table`, which builds the 3-D *forward* tables, uses the same construction and rule. Measured since (`measure_stack_table_cross_material`), the defect is there and larger than the $\sim 1\%$ first guessed. With a fast slab half a pitch from the planes, 10/pitch is off by up to 25% of the operator's scale, and that includes the reverberation block of a background plane next to the slab, not only cross-material pairs. 20/pitch converges to $\leq 3 \times 10^{-4}$. A no-slab control is exact. The correction is recorded in note (III). For the adjoint the remedy is simply the finer cutoff.

5 The Dietrich–Kormendi limit

At zero contrast, $\psi = \psi^{\text{inc}}$ and $\tilde{\psi} = \mathbf{R}^T \bar{r}$, the residual carried into the reference by reciprocity. Equation (6) becomes

$$\left. \frac{\partial M}{\partial m_{i,\ell}} \right|_{T_0=0} = \text{Re} \sum_{s,\omega,k_y} w_{ky} (\mathbf{R}^T \bar{r})_i^\top \left. \frac{\partial T_0^{(i)}}{\partial m_{i,\ell}} \right|_0 \psi_i^{\text{inc}}, \quad (10)$$

the reference Green's function to the receivers, times the Born vertex, times the reference field from the source. That is the Dietrich–Kormendi Fréchet kernel, $\delta R = \int G^\dagger \delta \mathbf{P} G^\downarrow dz$, contracted with the residual: a migration of the residual in the stratified reference, as Tarantola observed for the homogeneous case. So the D&K derivative is the first gradient of this inversion, and the full formulation adds exactly two things: the multiple scattering between voxels, through $(\mathbf{I} - G_0 T_0)^{-1}$, and the resummed intra-voxel scattering, through $\partial T_0 / \partial m$ away from zero.

This limit supplies the one *absolute* gate of the ladder (A3). The earlier invariants and self-consistency checks are all blind to an overall scale or sign; this one is not.

5.1 The limit, measured (gate A3)

The first draft planned the AD Kennett Jacobian of `Kennett_Reflectivity` as the arbiter. That code's PP reflectivity is referenced to a water layer above a fluid–solid seabed, so matching it would put the voxel side's sources and receivers in water, which the corrected layered kernel does not support. The gate instead uses the D&K kernel in its simplest exact form. In a uniform medium the Born PP reflection of a layer of thickness h , referenced at its top, is exactly

$$\delta R_{PP} = R_{\text{lin}}(\delta m) (1 - e^{2i\omega\eta_P h}), \quad (11)$$

with R_{lin} the Aki & Richards linearised PP coefficient. That closed form shares no code with the voxel side. On the voxel side, cubes of side $2a$ on a $2a \times 2a$ lattice form a sheet at mid-layer with Born source density $(2a)^{-2} \partial T_0 / \partial m|_0 \psi^{\text{inc}}$ per unit area. This uses the gradient code's own Richardson derivative. The sheet radiates through `vertical_kernel_9x9`, the kernel the receiver maps and the vertical sweep carry, and is projected onto up-going P by the mode bridge. P polarisations follow Aki & Richards on both sides. The sheet samples the layer at its midpoint and T_0 carries its form factor, so the two differ by $O(a^2)$. Any $O(1)$ defect (a sign, a missing 2π or $(2a)^{-2}$, a wrong vertex) would leave a flat residual.

Incidence	residual at $2a = 0.4$	residual at $2a = 0.05$ km	slope in a ($\Delta\lambda$, $\Delta\mu$, $\Delta\rho$)
normal	2.2×10^{-2}	3.3×10^{-4}	2.01, 2.01, 2.01
30° , in k_x	1.1×10^{-2}	1.7×10^{-4}	2.01, 2.01, 2.01
48.6° , split over k_x and k_y	3.4×10^{-3}	4.1×10^{-5}	1.99, 2.11, 1.98
<i>must fail</i> : vertex sign flipped	2.0	2.0	0.00
$(2a)^{-2}$ omitted	0.84	1.0	-0.08

The medium is $\alpha = 5$, $\beta = 3$, $\rho = 2.5$ at 1 Hz; the largest residual per row is quoted. The voxel Born vertex, the kernel’s absolute normalisation and the polarisation conventions are therefore right in absolute terms, not only consistent with one another. A3 does *not* cover the finite-lattice assembly or the adjoint chain (the unit tests of §8 check those against brute force). It also does not cover the lattice’s non-specular orders, which are evanescent for $2a$ below half a wavelength and are dropped, or a stratified reference.

6 Reference-layer parameters

Note (III) found that the crust gradient belongs in the reference, not in the scattering series. The reference parameters m^{ref} enter d^{ref} , $\mathbf{R}$, ψ^{inc} and G_0 . Differentiating (1)–(2) and using $\mathbf{R}T_0(\mathbf{I} - G_0T_0)^{-1} = \mathbf{R}(\mathbf{I} - T_0G_0)^{-1}T_0$,

$$\delta M = \text{Re} \sum \left[\bar{r}^\top \delta d^{\text{ref}} + \bar{r}^\top \delta \mathbf{R} \tau + \tilde{\psi}^\top T_0 \delta \psi^{\text{inc}} + (T_0^\top \tilde{\psi})^\top \delta G_0 \tau \right], \quad \tau = T_0 \psi. \quad (12)$$

The first term is the 1-D problem, already solved and gated. The second and third are the reference Green’s function differentiated at the receivers and the source. The fourth contracts the adjoint and forward contrast sources through $\partial G_0 / \partial m^{\text{ref}}$ and costs about one G_0 matvec per reference parameter, which is acceptable because G_0 depends only on the reference. Whether `GlobalMatrix.layered_greens` can be differentiated through the existing `gmm_torch` machinery has not been established. The recommended first scheme is *alternating*: update the reference with the existing 1-D machinery at fixed voxels, then the voxels with (6) at fixed reference. Joint inversion waits for (12).

7 Cost, Gauss–Newton, and place in the lineage

Per source, frequency and k_y node, the gradient costs one forward and one adjoint Krylov solve on the same operator (9), whatever the number of voxels or receivers. In the frequency domain the stored state is ψ alone, nine complex numbers per voxel, with none of the wavefield checkpointing a time-domain adjoint needs. A Fréchet-derivative product $F \delta m = \mathbf{R}(\mathbf{I} - T_0G_0)^{-1} \delta T_0 \psi$ is one further solve, so Gauss–Newton by inner conjugate gradients costs two solves per inner iteration, and the second-order Born Hessian hierarchy of `frechet_derivatives.tex` is the 1-D template for its approximations.

In the lineage of note (II), this is the distorted-Born iterative T-matrix inversion of Jakobsen & Ursin, with an analytic, resummed cell T-matrix and an exact stratified reference; and in the thesis’s own terms, the Schwinger–Levine functional already pairs the forward field with the $-k_y$ field, which is the pairing (6) uses.

8 Implementation

`cubic_scattering/sweep_gradient.py` implements (5)–(6) for one source, frequency and k_y node on the $2^{1/2}$ -D directional-sweep solver. The sum over k_y nodes, with the quadrature weight and phase of §3, is left to the caller.

Function	What it does, and the result it rests on
<code>misfit_and_gradient</code>	forward solve, data, misfit, then $\text{Re}(\mathbf{F}^H r)$ through <code>frechet_adjoint_product</code> ; refuses caches that are not at $\pm k_y$
<code>frechet_vector_product</code>	the tangent-linear $\mathbf{F} \delta m = \mathbf{R}(\mathbf{I} - T_0 G_0)^{-1} \delta T_0 \psi$ by one further forward solve; the Gauss–Newton building block
<code>frechet_adjoint_product</code>	$\text{Re}(\mathbf{F}^H q)$ for any data-space q : adjoint solve and local contraction (6)
<code>adjoint_solve</code>	the forward solver at $-k_y$ on $\mathbf{W}_\pm^{-1} \tilde{\mathbf{b}}$, then $\tilde{\psi} = \mathbf{W}_\pm \tilde{\psi}_-$ (A1)
<code>rayleigh_t0_and_derivative</code>	Rayleigh T_0 per voxel in its plane’s medium, and $\partial T_0 / \partial m$ by one Richardson step (A2)
<code>whole_space_receiver_map,</code> <code>layered_receiver_map</code>	a <i>dense</i> $\mathbf{R}$ (one row per recorded component), so $\mathbf{R}^T$ is exact by construction; the force form (7) (A1b) is the route to take only if $\mathbf{R}$ becomes too large to store

Only the Rayleigh cube is supported; the resonance composite fails the T_0 law. The numerical settings (GMRES tolerance, iteration cap, Richardson step) are required arguments with no defaults. Wiring the gradient to the ω annotations exposed that `compute_cube_tmatrix` and six helpers were annotated `omega: float` while every sweep gate passes a complex ω . The annotations were widened to `complex`; the runtime is unchanged.

Tests (`cubic_scattering/tests/test_sweep_gradient.py`, fourteen):

- the adjoint solve against a dense solve of $(\mathbf{I} - G_0^T T_0^T) \tilde{\psi} = \tilde{\mathbf{b}}$, to 10^{-9} ;
- the gradient against brute-force central differences of the misfit, to 10^{-6} . In a whole space this is at weak and moderate contrast. In the fast-slab crust it uses stratified forward and adjoint caches, per-plane media, and a receiver *inside* the slab;
- the receiver map against the entries of the dense sweep operator G_0 , with asymmetric offsets, to 10^{-12} ;
- on a uniform model, the layered receiver map reduces to the whole-space map to 10^{-9} at 3 Hz. At 0.3 Hz the two differ by the physical seabed return, 9.7×10^{-4} , flat in the quadrature rule and falling with frequency; the first version of that test ran at 0.3 Hz and failed for that reason;
- $\mathbf{F} \delta m$ against central differences of the predicted data, to 10^{-7} ; the dot-product identity, to 10^{-10} ; and the misfit gradient equal to $\text{Re}(\mathbf{F}^H r)$;
- $\partial T_0 / \partial m$ against an independent small-step difference; zero residual giving zero gradient; and three fail-fast diagnostics.

Deliberate defects were injected to show the tests can fail. Each of these turned at least one test red: $\mathbf{W}$ in place of $\mathbf{W}_\pm$; the adjoint solved at $+k_y$; the residual left unconjugated; $\partial T_0 / \partial m$ frozen at its Born value; the lateral phase sign of $\mathbf{R}$ flipped; and the multiple-scattering term dropped from $\mathbf{F} \delta m$.

These tests check the gradient of the *discrete* misfit at moderate contrast. They do not replace A3 (absolute agreement with the D&K limit), A4, or A5 (the Taylor test).

8.1 The dot product, measured (gate A4)

`gate_a4_dot_product` checks $\text{Re}(q^H \mathbf{F} \delta m) = \delta m \cdot \text{Re}(\mathbf{F}^H q)$ for five random pairs $(\delta m, q)$ per case. $\mathbf{F} \delta m$ is computed forward, and $\mathbf{F}^H q$ by the adjoint route the gradient uses. The two share only ψ and $\partial T_0 / \partial m$, and there is no step, so the identity holds to solver tolerance only if the adjoint really is the transpose. That covers the A1 and A1b laws, the receiver transpose, the

conjugate on q and the real part, all at once. GMRES runs at 10^{-12} , and the spectral radius $\text{spr}(G_0T_0)$ is measured for each case.

Case	contrast	$\text{spr}(G_0T_0)$	mismatch	<i>must fail</i> : Born F	<i>must fail</i> : $\mathbf{W}$ for $\mathbf{W}_\pm$
whole space, $ka_S = 0.26$	$1\times$	0.011	4.1×10^{-13}	3.2×10^{-2}	0.11
	$4\times$	0.041	4.4×10^{-13}	0.12	0.70
	$8\times$	0.075	2.4×10^{-11}	8.1	3.4
stress, $ka_S = 0.52$	$25\times$	0.427	7.6×10^{-12}	0.22	0.72
	$40\times$	0.804	6.6×10^{-12}	1.1	24
stratified, receiver in slab	$1\times$	0.056	1.8×10^{-13}	2.7×10^{-2}	5.0×10^{-2}
	$4\times$	0.196	1.1×10^{-12}	6.2×10^{-2}	0.15
	$8\times$	0.334	3.9×10^{-12}	9.3×10^{-2}	0.23

Contrast is a multiple of the project’s test contrast (2 GPa, 1 GPa, 0.1 g/cm³), perturbed per voxel. The identity holds to $\leq 2.4 \times 10^{-11}$ everywhere, including $\text{spr} = 0.80$, and both calibration legs fail in every row.

How strong the multiple scattering can get. With *physical* Rayleigh T_0 inside its validity ($ka_S \leq 0.3$) and the project’s contrasts, $\text{spr}(G_0T_0)$ stays at about 0.3 or below. Reaching $\text{spr} \approx 0.8$ needs $ka_S \approx 0.5$ and $\Delta\mu \approx 1.8\mu_0$, outside the Rayleigh cube’s validity. The dot product is an algebraic identity, so the *stress* rows are run there, labelled as such, to exercise the resolvent. The $\text{spr} = 0.92$ of the rung-5 solver gate used random T_0 blocks, not physical ones, so the ladder’s earlier plan to run A5 “at the rung-5 spectral radius” with physical T_0 was not achievable as written.

8.2 The Taylor test, measured (gate A5)

A4 shows the adjoint is the transpose of the tangent-linear model. The Taylor test shows that model is the derivative of the *misfit*. For a random direction δm ,

$$\text{Rem}_2(\varepsilon) = \mathbf{M}(m + \varepsilon \delta m) - \mathbf{M}(m) - \varepsilon \nabla \mathbf{M} \cdot \delta m = O(\varepsilon^2) \quad (13)$$

if and only if $\nabla \mathbf{M}$ is right. Any gradient error leaves an $O(\varepsilon)$ piece, and $\text{Rem}_2/\varepsilon^2$ must settle to $\frac{1}{2} \delta m^\top (\nabla^2 \mathbf{M}) \delta m$. `gate_a5_taylor` evaluates $\mathbf{M}$ with the full nonlinear model at every ε : T_0 recomputed from the perturbed contrasts, the Foldy–Lax solve and the receiver map. It uses data from a different “true” model, so the residual is $O(1)$. ε runs from 10^{-1} to $10^{-1}/2^{10}$ of the base contrast.

Case	spr	slope, Rem_2	steady to	Rem_1	Born $\nabla \mathbf{M}$
whole space, $ka_S = 0.26$, $8\times$	0.071	2.000	3.4×10^{-6}	0.993	1.021
stratified, receiver in slab, $8\times$	0.358	2.004	7.5×10^{-5}	1.001	1.007
stress, $ka_S = 0.52$, $40\times$	0.739	2.012	1.9×10^{-4}	0.988	1.005

Columns: spr is $\text{spr}(G_0T_0)$; “steady to” is the relative change of $\text{Rem}_2/\varepsilon^2$ over the finest three ε ; the last two are the slopes of the calibration legs, which must be ≈ 1 . Rem_1 omits the gradient term, and the Born-only gradient back-propagates the residual without the adjoint solve. Their slopes are asymptotic, fitted over the finest three ε . No solver floor appears: $\text{Rem}_2/\varepsilon^2$ is steady to four digits at $\varepsilon \approx 10^{-4}$ with GMRES at 10^{-12} .

Two design errors in the first run, neither a gradient defect, are recorded because they would recur. The calibration slopes were first fitted over the whole range, and the gate reported itself blind. Where $\nabla \mathbf{M} \cdot \delta m$ is small beside the curvature (Rem_1 even changes sign), or the Born error is small ($\text{spr} \approx 0.07$), the $O(\varepsilon^2)$ term dominates at large ε , and only the small- ε end shows the $O(\varepsilon)$ defect the legs exist to expose. In the stress case $\nabla \mathbf{M} \cdot \delta m \approx -0.04$ against a curvature of 2.6, so Rem_1 turns linear only for $\varepsilon \ll 0.017$, and the first ε grid, stopping at $10^{-1}/2^6$, did not reach it.

9 Validation ladder

In order. Each rung is a prerequisite for trusting the next. Following note (III), every invariant is paired with at least one absolute comparison, because reciprocity and dot-product tests are homogeneous of degree one and blind to an overall scale error.

	Claim	Arbiter	Status
A0	$T_0^\top = \mathbf{W}T_0\mathbf{W}^{-1} = \mathbf{W}_\pm T_0\mathbf{W}_\pm^{-1}$	direct measurement, with failing calibration leg	measured g:t0_reciprocity
A1	$G_0(k_y)^\top = \mathbf{W}_\pm G_0(-k_y)\mathbf{W}_\pm^{-1}$ on the assembled sweep operator, and identity (9)	dense materialisation at $\pm k_y$, three failing calibration legs	measured g:a1_sweep_reciprocity
A1b	$G(r \leftarrow v)^\top = \mathbf{W}_\pm G(v \leftarrow r)\mathbf{W}_\pm^{-1}$ and form (7) of $\mathbf{R}^\top$	k -domain kernel; real-space dense $\mathbf{R}$, two failing calibration legs	measured g:a1b_receiver_transpose
A2	$\partial T_0/\partial m$, and its Born limit	Richardson differences; directional consistency; Born weights as $ka \rightarrow 0$; imaginary step must fail	measured g:a2_t0_derivative
A3	D&K limit (10)	exact Born thin layer (11) (absolute; slope 2 in pitch; two failing calibration legs)	measured g:a3_dk_limit
A4	$\text{Re}(q^\text{H}\mathbf{F}\delta m) = \delta m \cdot \text{Re}(\mathbf{F}^\text{H}q)$	dot-product test, $\mathbf{F}$ forward vs $\mathbf{F}^\text{H}$ adjoint, to spr = 0.80; two failing calibration legs	measured g:a4_dot_product
A5	gradient (6) at finite contrast	Taylor test (13), slope 2, to spr = 0.74 (labelled stress case); two calibration legs at slope 1	measured g:a5_taylor

Every rung has passed. The two that exercise the multiple scattering, A4 and A5, reach spr ≈ 0.36 with physical T_0 and spr ≈ 0.74 – 0.80 in a labelled stress case outside the Rayleigh cube’s validity (§8.1).

10 Status

Item	Status	Evidence
Gradient identity (5)–(6)	exact algebra	§3
Adjoint = forward at $-k_y$ (9)	measured , $\leq 3 \times 10^{-16}$	A1
T_0 half of (8), Rayleigh cube	measured , exact	g: t0_reciprocity
T_0 half, resonance composite	fails , 1.6–17%	g: t0_reciprocity
G_0 half, assembled, whole space and stratified	measured , $\leq 2 \times 10^{-14}$	A1
Receiver transpose $\mathbf{R}^\top$, form (7)	measured , $\leq 10^{-9}$ at $k_r^{\max} = 20/\text{pitch}$	A1b
Transverse rule 10/pitch, cross-material planes	inadequate , 1.4×10^{-2}	A1b
$\partial T_0/\partial m$, Richardson differences	measured , $\leq 2.5 \times 10^{-11}$	A2
T_0 holomorphic in m (complex step)	no ; off by 4×10^{-4} – 2×10^{-2}	A2
D&K limit, uniform reference	measured , $O(a^2)$ to $\leq 3.3 \times 10^{-4}$	A3
Dot product, F forward vs $\mathbf{F}^H$ adjoint	measured , $\leq 2.4 \times 10^{-11}$ to $\text{spr} = 0.80$	A4
$\text{spr}(G_0 T_0)$, physical Rayleigh T_0 within validity	$\lesssim 0.36$	A4, A5
Taylor test of ∇M against the nonlinear misfit	measured , slope 2.000–2.012 to $\text{spr} = 0.74$	A5
Reference-layer gradient	1-D term gated elsewhere; coupling terms unvalidated	§6
Gradient code, 2½-D, one k_y node	built; passes A0–A5	§8

The validation ladder is complete. A0–A2 measure every ingredient of the gradient: the adjoint as the forward solve at $-k_y$ conjugated by $\mathbf{W}_\pm$, the receiver transpose, and a Richardson derivative of T_0 . The code (§8) assembles them. A3 confirms its Born limit against exact physics in absolute terms. A4 shows the adjoint is the exact transpose of the tangent-linear model, and A5 that this model is the derivative of the nonlinear misfit, both into strong multiple scattering.

What the ladder does not cover is equally specific:

- the sum over k_y nodes, which is the caller’s (§3);
- the 3-D solver, and reference-layer parameters beyond the 1-D term (§6);
- T_0 other than the Rayleigh cube: the resonance composite fails the law, and the Galerkin T_0 is untested;
- physical T_0 at spr above about 0.36, which lies outside the Rayleigh cube’s validity;
- the stratified A3.

The A1 and A1b evidence is a few voxels at one frequency per background: small, but the laws are pairwise identities, so lattice size does not enter them. A2 covers the Rayleigh cube only; the resonance composite is excluded (§4), and the 27-component Galerkin T_0 is untested. A1b also found a defect outside the adjoint: the transverse rule is under-resolved for planes near a material contrast, which reaches the *forward* 3-D tables at up to 25% of the operator’s scale (§4.1).

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The measured values in this note are reproducible from `scripts/gate_t0_reciprocity.py`, `scripts/gate_a1_sweep_reciprocity.py`, `scripts/gate_a1b_receiver_transpose.py`, `scripts/gate_a2_t0_derivative.py`, `scripts/gate_a3_dk_limit.py`, `scripts/gate_a4_dot_product.py` and `scripts/gate_a5_taylor.py`. Values quoted from other notes are reproducible from the gates those notes cite.